Mobility-Aware UAV Positioning and Secret Key Rate Optimization for HAP–UAV CV-QKD

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***Abstract*—This provisional internal-review study examines source selection and UAV positioning for a cloud-affected HAP–UAV discrete-modulated CV-QKD model. A finite catalog of 233 exact-source ensembles defines a common location-dependent numerical rate map. An offline stationary benchmark, a current-slot Greedy policy, and finite-horizon dynamic programming (DP) are compared under identical cloud realizations and speed constraints. Across 100 paired missions, expanding the candidate set from 10 to 100 raises the mean DP penalized objective by 11.41%. At blockage probability 0.8, mean zero-rate unavailability is 24.60%, 16.53%, and 4.07% for Stationary, Greedy, and DP, respectively. At the representative penalty, DP reduces travel by 88.37% relative to zero penalty while reducing mean usable rate by 0.515%. These are conditional manuscript-model outputs: the recorded two-dimensional mutual information and homodyne Holevo expression do not specify a consistent Bob measurement. The results characterize the saved numerical optimization and do not establish physical quantum security.**

***Index Terms*—Continuous-variable quantum key distribution, UAV, high-altitude platform, cloud blockage, probabilistic shaping, dynamic programming.**

Internal review only. Security-model correction and author review remain pending. Status:
`UNVALIDATED_SECURITY_MODEL`;
`NOT_READY_FOR_PUBLICATION_SCALE_RUNS`.

I. HAP–UAV CV-QKD System and Problem Formulation

A. Mobility-Aware UAV Positioning and SKR Optimization

We consider one UAV at a fixed altitude over a $1\,\mathrm{km}\times 1\,\mathrm{km}$ area, discretized into J candidate positions $\mathcal{Q}=\{\mathbf{q}_j\}_{j=1}^{J}$. The mission has N_t slots of duration Δt. Let $c_{j,n}\in\{0,1\}$ indicate cloud blockage (one means blocked). The matrix $\mathbf{C}=[c_{j,n}]$ contains spatially and temporally correlated Bernoulli states. P2 and P3 know the complete matrix and are offline benchmarks; Greedy observes only the current slot.

For an unblocked link, the recorded manuscript-style raw rate is

$$K_j(\nu,V_A)=\beta_{\mathrm{rec}}I^{2D}_{AB,j}(\nu,V_A)-\chi_{BE,j}(\nu,V_A). \tag{1}$$

Equation (1) is retained as a conditional numerical metric, subject to the measurement-consistency limitations below.

B. Recorded Physical-Layer Model and Security Boundary

Alice sends the coherent-state ensemble $\{p_k,\alpha_k\}_{k=1}^{256}$. With $[\hat{x},\hat{p}]=2i$ and vacuum quadrature variance one, the normalization is $V_A=2\sum_k p_k|\alpha_k|^2=2\bar{n}$. The source moments C_s,w_s are computed from this same ensemble; the subscript distinguishes them from the cloud matrix. For clear-sky power transmittance η, the saved evaluator uses

$$\begin{aligned}\eta_f&=\langle\sqrt{\eta}\rangle^2,\\ N=\eta_f\xi_f&=(\langle\eta\rangle-\eta_f)V_A+\xi_{\mathrm{ch}}\langle\eta\rangle,\\ V&=1+V_A,\qquad W=1+\eta_f V_A+N,\\ Z&=2\sqrt{\eta_f}C_s-\sqrt{2Nw_s}.\end{aligned}$$

The convention is $\xi_n=\xi_{\mathrm{ch}}$. The additive N is output-referred excess noise; loss-induced vacuum is already included in W's unit baseline. The active detector is ideal (efficiency one, electronic noise zero). No imperfect trusted-detector claim is made. The clear-sky distribution retains a separate angle-of-arrival outage atom at $\eta=0$ and its active density on $[0,1]$, without renormalizing. Moment reduction at the covariance level has a precedent for Gaussian modulation [1]; it is not an identity for the full discrete-input fading likelihood.

Specifically, the saved MI kernel uses the normalized likelihood

$$p_{\mathrm{num}}(\mathbf{y}|k)=\frac{\exp[-\|\mathbf{y}-2\sqrt{\eta_f}(\Re\alpha_k,\Im\alpha_k)\|^2/(2(1+N))]}{2\pi(1+N)}$$

and integrates both real components to obtain I^{2D}_{AB}. The Holevo branch uses the standard-form covariance with entries V,W,Z and the *homodyne* conditional eigenvalue

$$\nu_3^{\mathrm{hom}}=\sqrt{V(V-Z^2/W)}.$$

These two branches do not specify one consistent single-mode Bob measurement: a homodyne outcome has one quadrature, while joint measurement of both quadratures adds measurement vacuum. The measurement-dependent Holevo expressions are explicitly different in [2]. Consequently, all reported "usable SKR" values are conditional outputs of the recorded manuscript model, not established achievable secret-key rates.

Further, evaluating only Z_- does not establish the worst-case Holevo value over all admissible correlations; the general interval-maximization caveat in [2, Sec. 4, footnotes 11–12] also applies. A consistent measurement/noise specification, admissible correlation treatment, parameter estimation, and fading/outage conditioning protocol require author review. No correction is applied to the historical numerical results in this manuscript revision.

II. Optimization Problem

At each location, the finite exact-source catalog $\mathcal{G}$ contains 197 MB-shaped and 36 uniform 256-QAM ensembles. Searching the MB shaping parameter ν and modulation variance V_A gives

$$\mathcal{P}_1: \quad K_j^\star = \left[\max_{(\nu,V_A)\in\mathcal{G}} K_j(\nu, V_A)\right]^+. \tag{2}$$

Each candidate uses its own authenticated source moments; the raw maximum is selected before clipping. Channel moments independent of the source can be shared, while the modulation-dependent fading noise is evaluated for each V_A. The optimum is restricted to $\mathcal{G}$, rather than a continuous shaping domain. With complete cloud blockage, $S_{j,n} = (1 - c_{j,n})K_j^\star$. Offline stationary selection is

$$\mathcal{P}_2: \quad j_{\text{fix}}^\star = \arg\max_j \frac{1}{N_t}\sum_{n=1}^{N_t} S_{j,n}. \tag{3}$$

The chosen location balances clear-sky rate and the full cloud sequence; it need not be the location with the largest clear-sky rate alone. For $j_n \in \{1, \ldots, J\}$ and $d_{i,j} = \|\mathbf{q}_i - \mathbf{q}_j\|_2$, mobility optimization is

$$\mathcal{P}_3: \quad \max_{\{j_n\}} \quad \underbrace{\frac{1}{N_t}\sum_{n=1}^{N_t} S_{j_n,n}}_{\overline{K}} - \frac{\lambda}{D_{\text{ref}}} \underbrace{\sum_{n=2}^{N_t} d_{j_{n-1},j_n}}_{D} \tag{4}$$
$$\text{s.t.} \quad d_{j_{n-1},j_n} \le v_{\max}\Delta t, \quad n \ge 2.$$

The objective is $U = \overline{K} - \lambda D/D_{\text{ref}}$, with $D_{\text{ref}} > 0$ and $\lambda \ge 0$ in bit/use. Staying and a free initial position are allowed; no terminal, deployment, or return requirement is imposed. Distance is a mobility-cost surrogate, not propulsion energy.

III. Proposed Solution

P1 exhausts the finite catalog once per clear-sky position, reusing exact source moments; P2 compares all full-horizon location means. For P3, let $\mathcal{N}(j) = \{i : d_{i,j} \le v_{\max}\Delta t\}$. DP uses

$$F_1(j) = S_{j,1}, \tag{5}$$
$$F_n(j) = S_{j,n} + \max_{i\in\mathcal{N}(j)} \left[F_{n-1}(i) - \frac{N_t\lambda}{D_{\text{ref}}} d_{i,j}\right]. \tag{6}$$

Every feasible predecessor is considered, including staying, and the retained state summarizes the best prefix ending at that location. Backtracking saved predecessors from $\arg\max_j F_{N_t}(j)$ recovers the finite-grid optimum, with $U^\star = \max_j F_{N_t}(j)/N_t$ and time $O(N_t J^2)$. Backpointers require $O(N_t J)$ storage, in addition to the $O(J^2)$ distance matrix. The stored implementation state F_n/N_t has reward $S_{j,n}/N_t$ and edge cost $\lambda d_{i,j}/D_{\text{ref}}$, which is the same normalization. Both P2 and DP know the complete cloud sequence offline. Greedy starts at $\arg\max_j S_{j,1}$, then maximizes the current-slot score $S_{j,n} - N_t\lambda d_{j_{n-1},j}/D_{\text{ref}}$ over feasible moves, in $O(N_t J)$ time. All methods share rates, clouds, and constraints. DP dominates their penalized objectives, not necessarily each mission's SKR or availability. Their difference also reflects DP's future-state and initial-position information, so it does not measure an online control gain. Without feasible movement, DP reduces to P2. P2 itself is independent of the movement penalty because its path has zero distance.

IV. Numerical Results and Discussion

The synthetic HAP/UAV altitudes are 20/1 km; the $1\,\text{km}^2$ grid has 100 centers spaced 100 m apart. Missions have 30×20 s slots, $v_{\max} = 15$ m/s, and $D_{\text{ref}} = 1000$ m. The synthetic deployment places the HAP horizontally at $(11419.6551, 450)$ m and uses UAV coordinates $50, 150, \ldots, 950$ m along each axis. Saved missions use seed 20261015 and realization IDs 0–99. The retained optical settings are 1550 nm, full divergence 233 μrad, receiver diameter 0.08 m, $\beta_{\text{rec}} = 0.95$, $\xi_{\text{ch}} = 0.0186$ SNU, and ideal detection. Saved Gaussian-copula complete-blockage clouds default to $(p, \ell, \rho) = (0.4, 250\,\text{m}, 0.9)$, with latent spatial covariance $\Sigma_{ij} = \exp(-d_{i,j}/\ell)$ and a stationary AR(1) coefficient ρ. Latent parameters differ from Bernoulli-cloud correlations. Each latent state is thresholded at $\Phi^{-1}(p)$, where Φ is the standard-normal distribution function; the marginal blockage probability is prescribed, rather than measured meteorology. We compare 100 paired missions at fixed $\lambda_{\text{ref}} = 9.468026677248516 \times 10^{-6}$ bit/use. Means and pointwise confidence intervals use independent whole missions, not cells or slots; methods share each realization ID and configurations use common innovations. No new physical-layer evaluation is required. These intervals describe mission variability, rather than finite-key parameter-estimation uncertainty or a quantum-security bound. Availability means $S > 0$, without an engineering threshold. In Fig. 1(a), expanding deterministic nested position sets from $J = 10$ to 100 raises mean DP U by 11.409% relative to $J = 10$. The improvement combines more spatial choices and greater connectivity; optimal P2/DP objectives cannot decrease as the feasible set expands. The comparative presentation follows [3].

At $p = 0.8$, mean unavailability is 24.60%, 16.53%, and 4.07% for P2, Greedy, and DP, respectively [Fig. 1(b)]. Cloud-aware movement improves this observed metric, although P3 optimizes U. Increasing p under nested masks removes rewards, so optimal P2/DP objectives cannot increase; no general availability ordering follows. As latent ρ increases from 0 to 0.98, DP distance decreases from 1038.25 to 31.84 m [Fig. 1(c)], while its mean objective gain over P2 decreases from 23.712% to 0.155%, relative to the corresponding P2 mean. Persistent favorable locations make offline stationary selection competitive. At the high-correlation endpoint, the latent persistence time exceeds the mission horizon, allowing P2 to exploit locations favorable throughout the mission. This is not a universal monotonic law or a demonstration with forecast errors.

On the same default missions, changing DP from zero penalty to λ_{ref} reduces mean distance by 88.370% and mean usable SKR by 0.515%, relative to their respective zero-penalty means. Thus, most movement can be avoided with a small numerical rate loss. P1 selects MB $(\nu, V_A) = (0.125, 8)$

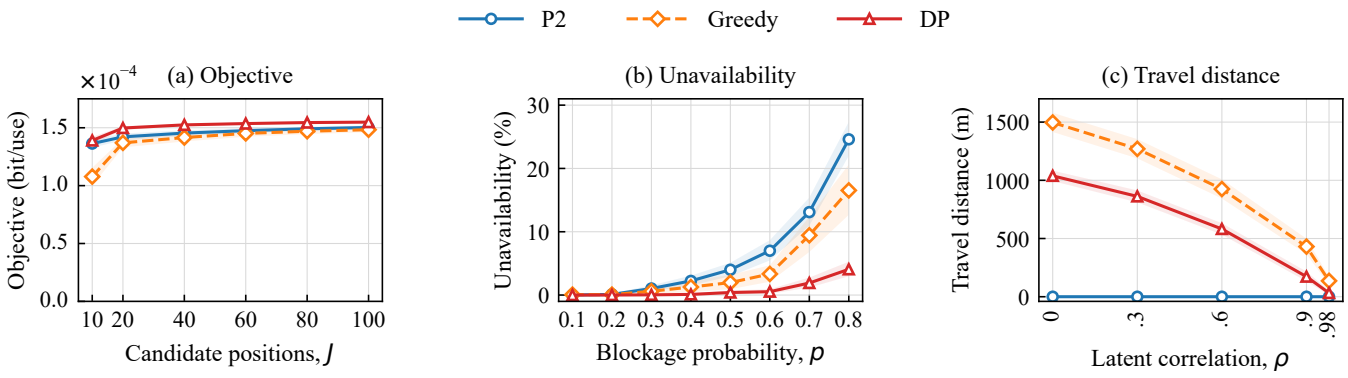


Fig. 1. Parameter sensitivity at λ_{ref}: (a) objective versus candidate count; (b) zero-SKR unavailability versus blockage probability; (c) travel distance versus latent temporal correlation. Curves use 100 paired missions; shaded bands are pointwise 95% whole-mission bootstrap intervals.

everywhere, at the MB catalog's variance boundary; positioning gains do not isolate shaping gain. All rates remain conditional outputs of the mixed 2D-MI/homodyne-Holevo model in Section I-B, without a validated quantum-security guarantee. Finite-key estimation, key aggregation across fading states, tracking interruptions, and reacquisition overhead are outside this comparison. Trusted receiver imperfections would require a separate model [4].

V. Conclusion

Finite-catalog source selection, offline stationary positioning, and DP trajectory optimization jointly characterize cloud-aware HAP–UAV performance. Under the evaluated configurations, DP improves the mobility-penalized objective over Stationary and Greedy, with observed availability benefits. A distance penalty substantially reduces movement while preserving most usable rate. Conclusions remain limited to a finite grid, synthetic clouds, offline future information, and an uncalibrated mobility-cost surrogate. The inconsistent MI/Holevo measurement pairing requires correction before interpreting numerical rates as secure keys.

References

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